

Coffee Sales Analytics and Customer Purchase Prediction

An End-to-End Data Science Project Using Excel, SQL, Python, and Machine Learning

Valerie Komen

Technical Report

July 2026

Repository: <https://github.com/Valerie-komen/Coffee-Sales-Analytics-ML>

Table of Contents

Table of Contents	2
1. Abstract	3
2. Introduction	3
2.1 Background	3
2.2 Problem Statement	3
2.3 Objectives	3
2.4 Research Questions	4
3. Dataset	4
4. Data Cleaning (Excel + SQL + Python)	4
4.1 Excel	4
4.2 SQL	5
4.3 Python	5
5. Exploratory Data Analysis	5
5.1 Which coffee sells the most — and is it the most profitable?	5
5.2 Who spends the most, and which country buys the most premium coffee?	6
5.3 Does loyalty-card membership increase spending?	6
5.4 Seasonality	7
5.5 Correlation structure and order-value distribution by roast	7
5.6 Bag-size revenue concentration	8
6. Machine Learning	8
6.1 Customer Segmentation (K-Means) — flagship result	8
6.2 Repeat-Purchase Classification — reported honestly	9
7. Interactive Dashboard	11
8. Results Summary	12
9. Limitations	13
10. Future Work	13
11. Responsible AI	13
11.1 Data Privacy	13
11.2 Bias and Fairness	14
11.3 Transparency	14
11.4 Responsible Deployment	14
12. Conclusion	14

1. Abstract

Customer transaction data can provide valuable insights into purchasing behavior, product performance, and business growth, yet it is often used only for descriptive reporting. This project analyzes coffee sales data from 2019 to 2022 to identify sales trends, customer purchasing behaviour, and product performance.

The dataset contains 1,000 transactions from 913 customers across the United States, the United Kingdom, and Ireland. The data was cleaned and validated using Excel, analyzed with SQL and Python, and presented through an interactive Plotly dashboard. Exploratory analysis was used to examine revenue, profitability, customer retention, product performance, and seasonal demand. Machine learning techniques were then applied to segment customers using K-Means clustering and to predict repeat purchases using supervised classification models.

The analysis found that Excelsa generated the highest revenue, while Liberica achieved the highest profit margin. More than half of all revenue came from the 2.5 kg product size, and only 2.7% of customers made a repeat purchase. Customer segmentation identified three distinct purchasing groups, providing meaningful insights for targeted marketing. In contrast, the repeat purchase prediction models performed only marginally better than random guessing, indicating that the available data contain insufficient predictive signal for reliable customer retention modeling.

The findings show that valuable business insights can be obtained through exploratory analysis and customer segmentation, while also highlighting the importance of evaluating whether a dataset is suitable for predictive modeling before deploying machine learning solutions.

2. Introduction

2.1 Background

Small and mid-sized retailers increasingly capture order-level transactional data, but most stop at descriptive reporting that describes what happened without informing what to do next. A more complete analytics practice combines descriptive analysis (what happened), diagnostic analysis (why), and predictive analysis (what is likely to happen) to generate actionable business insights from transactional data.

2.2 Problem Statement

Coffee retailers generate large volumes of sales data every day, but this information is often underutilized. Many businesses rely on basic reports and dashboards that describe past sales without providing deeper insights into customer behavior, product performance, or future purchasing patterns. As a result, opportunities to improve customer retention, optimize inventory, and make informed business decisions may be overlooked.

The coffee sales dataset used in this project contains valuable information on customers, products, and transactions collected between 2019 and 2022. However, without a structured analysis, it is difficult to identify meaningful trends, understand customer segments, or assess whether the available data can reliably support predictive decision-making. Addressing this gap is important for helping businesses move from reporting historical performance to making evidence-based decisions.

2.3 Objectives

- Build a reproducible data cleaning pipeline from raw Excel data to an analysis-ready database.
- Analyze sales performance, customer behavior, and product trends using SQL and Python
- Identify customer segments using clustering techniques.
- Develop and evaluate a machine learning model to predict repeat purchases.
- Present findings through an interactive dashboard and technical report.

2.4 Research Questions

- Which coffee type and roast generate the highest revenue and profit?
- Which customer characteristics, if any, predict repeat purchases?
- Can customers be grouped into meaningful purchasing segments?
- Does the loyalty-card program increase customer spending?
- What seasonal and country-level patterns exist in sales?

3. Dataset

Source: coffee_orders_dashboard.xlsx (raw_data.xlsx in this repository), a transactional export covering January 2, 2019 to August 19, 2022, with three sheets: orders, customers, and products.

Table	Rows	Key fields
orders	1,000 order lines	Order ID, Order Date, Customer ID, Product ID, Quantity, Country, Coffee Type/Name, Roast Type/Name, Size, Unit Price, Sales, Loyalty Card
customers	1,000 (913 unique)	Customer ID, Country, Loyalty Card (PII columns dropped — see Section 4)
products	48 SKUs	Product ID, Coffee Type, Roast Type, Size, Unit Price, Price per 100g, Profit

The dataset contains four coffee types (Arabica, Excelsa, Liberica, Robusta), three roasts (Light, Medium, Dark), and four bag sizes (0.2kg, 0.5kg, 1kg, 2.5kg) sold to customers based in the United States, United Kingdom, and Ireland.

4. Data Cleaning (Excel + SQL + Python)

Full, executed code is in notebooks/01_data_cleaning.ipynb and sql/cleaning_queries.sql; this section summarizes what was checked and found.

4.1 Excel

- Checked the dataset for missing headers, duplicate rows, and formatting issues.
- Confirmed date columns were true datetime values (not text) before export.
- Used PivotTable/lookup structure to validate summary statistics and compare results with Python analysis.

4.2 SQL

A normalized SQLite database was created with separate **orders**, **customers**, and **products** tables linked by foreign keys.

Data quality checks confirmed:

- No missing values in required fields.
- No duplicate records.
- All orders were linked to valid customers and products.
- Sales values matched the calculated **Quantity × Unit Price** for all transactions.

SQL queries were then used to analyze revenue, profit, customer spending, repeat purchases, loyalty program performance, sales trends, and country-level performance.

4.3 Python

Python was used for data cleaning, exploratory analysis, visualization, and machine learning. Personally identifiable information, including customer names, email addresses, phone numbers, and postal addresses, was removed before analysis, leaving only anonymous customer IDs.

The analysis was performed using **pandas** for data processing, **scikit-learn** for machine learning, **Matplotlib** and **Plotly** for visualization, and **SQLite** for database queries.

5. Exploratory Data Analysis

Full code in notebooks/02_exploratory_analysis.ipynb.

5.1 Which coffee sells the most — and is it the most profitable?

Coffee type	Revenue	Profit	Margin
Excelsa	\$12,306.44	\$1,353.71	11.0%
Liberica	\$12,054.08	\$1,567.03	13.0%
Arabica	\$11,768.50	\$1,059.16	9.0%
Robusta	\$9,005.24	\$540.31	6.0%

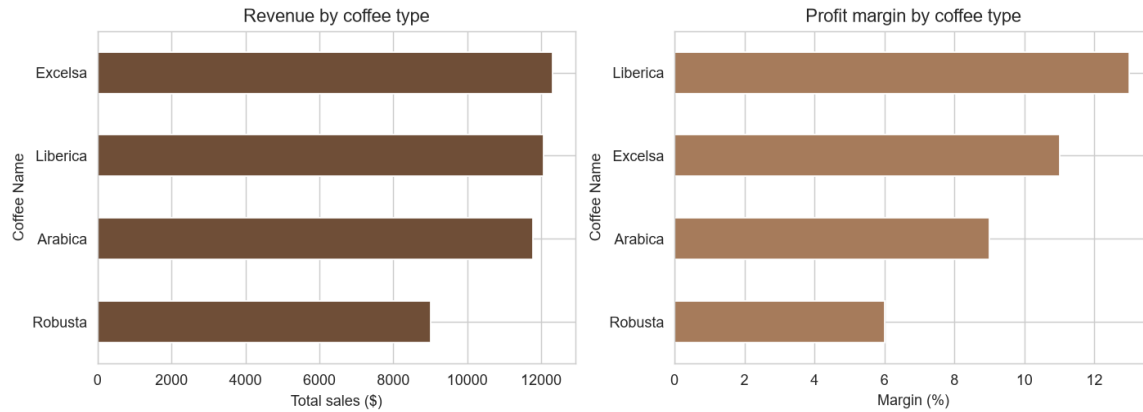


Figure 1. Revenue by coffee type (left) vs. profit margin by coffee type (right).

Excelsa was the top-performing coffee by revenue, while Liberica achieved the highest profit margin (13%). This indicates that revenue alone does not reflect overall profitability.

5.2 Who spends the most, and which country buys the most premium coffee?

Although the average purchase price was almost identical in the United States and Ireland (about \$13), the United Kingdom had the lowest average purchase price and the lowest revenue per customer. This suggests that the UK market may benefit from stronger marketing or customer engagement strategies rather than having inherently lower demand.

Country	Revenue	Share of total	Avg. spend / customer
United States	\$35,638.88	79.0%	\$50.05
Ireland	\$6,696.86	14.8%	\$48.53
United Kingdom	\$2,798.51	6.2%	\$44.42

5.3 Does loyalty-card membership increase spending?

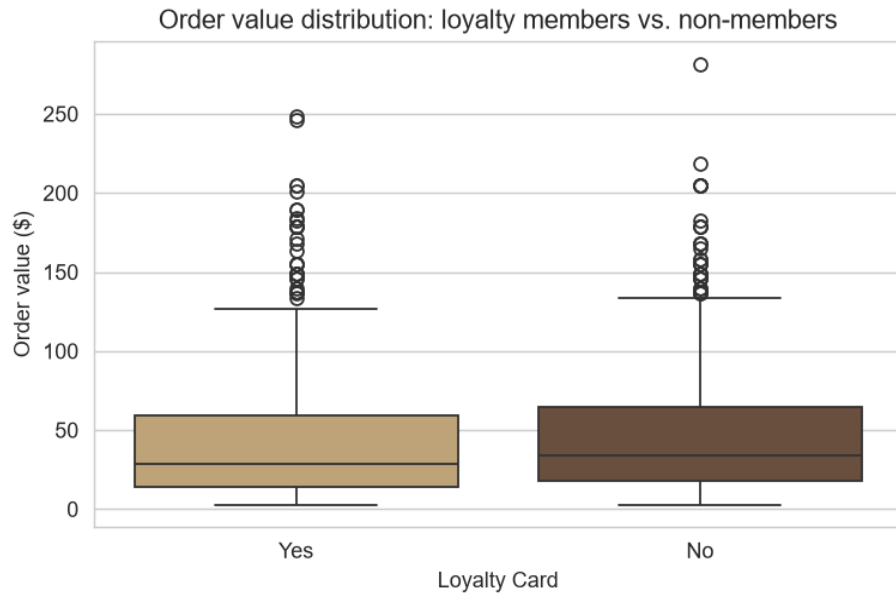


Figure 2. Order-value distribution: loyalty members vs. non-members.

No. Loyalty card holders' mean order value (\$45.08) and median (\$28.60) are both slightly below non-members' (\$49.12 mean, \$33.75 median), despite ~49% enrollment. The loyalty program has little impact on customer spending.

5.4 Seasonality

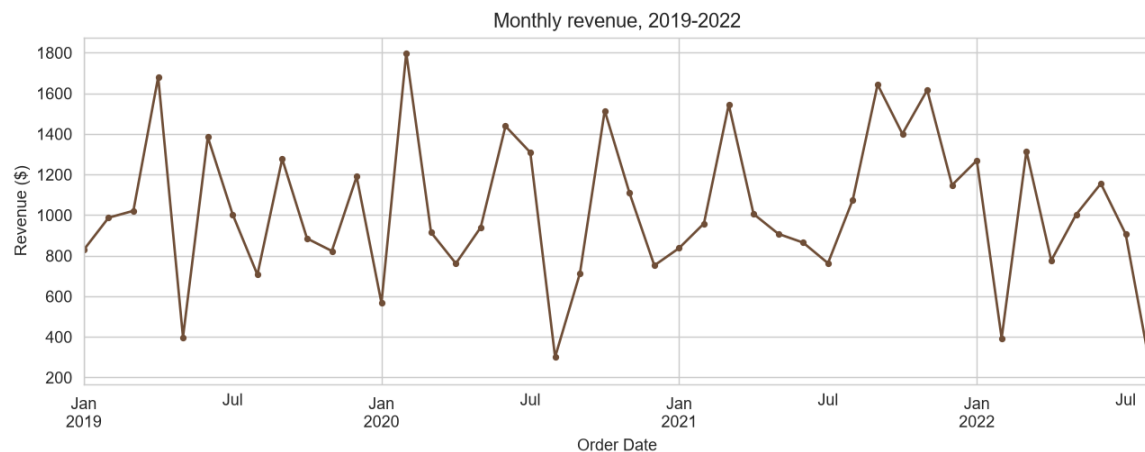


Figure 3. Monthly revenue, 2019–2022.

Sales peak in June (\$4,843 total across years) and drop in August (\$2,327 total) each year. Annual revenue was roughly flat 2019→2020 (\$12,187 → \$12,118, -0.6%), then grew 13.6% in 2021 (\$13,766). Based on the sales through August, total sales for 2022 are estimated at approximately \$10,595. This is lower than 2021, indicating a possible slowdown in sales that should be monitored

5.5 Correlation structure and order-value distribution by roast

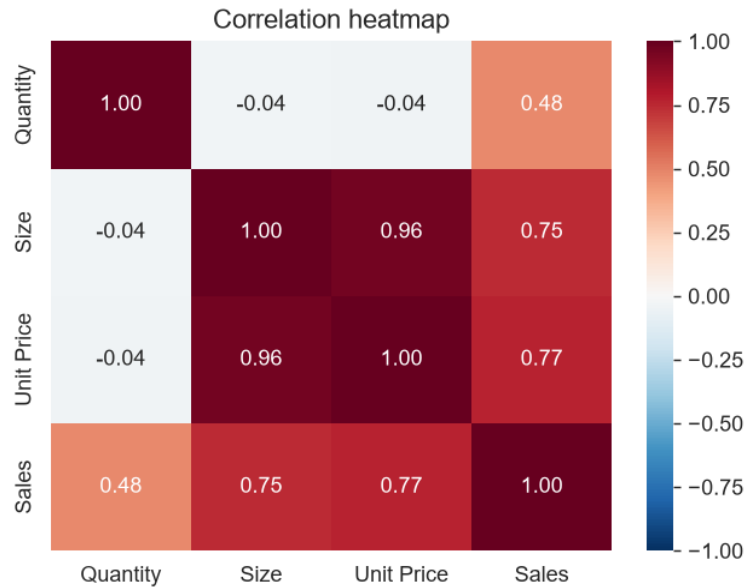


Figure 4. Correlation heatmap of numeric order features.

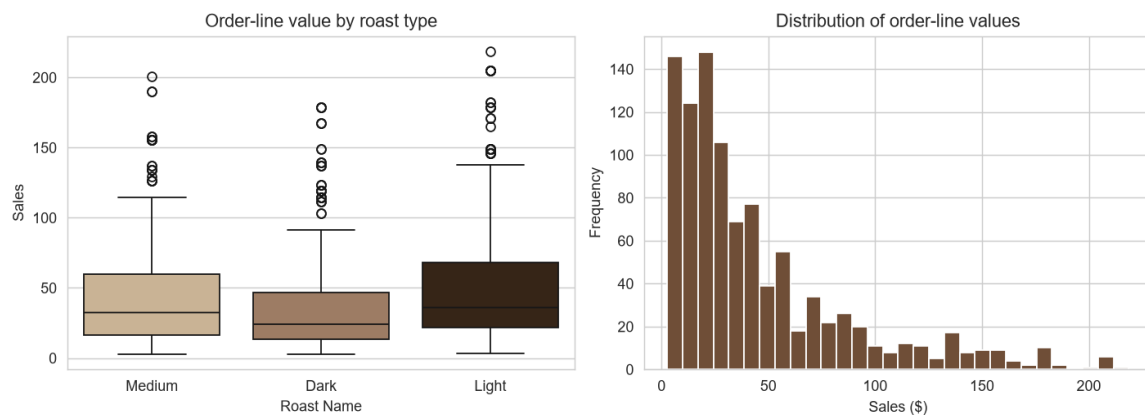


Figure 5. Order-line value by roast type (left) and overall distribution (right).

Sales correlates strongly with Size and Unit Price rather than the number of items ordered. Light roast has the highest average order value (\$52.12), followed by Medium (\$43.71) and Dark (\$39.58).

5.6 Bag-size revenue concentration

The 2.5 kg coffee size generated over half of all sales (52.7%), making it the biggest contributor to revenue. The 1 kg, 0.5 kg, and 0.2 kg sizes made up the remaining 47.3% combined.

6. Machine Learning

Full code and real, executed output in notebooks/03_machine_learning.ipynb. Two tasks are attempted, in order of how well the dataset actually supports them.

6.1 Customer Segmentation (K-Means) — flagship result

Customer groups were created using six features: how recently they made a purchase, how often they bought coffee, how much they spent, their average order value, average price paid, and how often they bought the 2.5 kg coffee size. These features were standardized before clustering to ensure they were on the same scale.

Different numbers of customer groups (2–6) were tested using the silhouette score. Three groups produced the best result (0.555), indicating the clearest separation between customers. The groups were then named based on their actual purchasing behaviour.

Segment	Customers	Share	Avg. spend	Avg. unit price	% bulk	% repeat
Occasional / low-spend buyers	671	73.5%	\$29.75	\$7.84	0%	1%
Bulk buyers	229	25.1%	\$98.77	\$28.06*	98%	2%
High-value / repeat buyers	13	1.4%	\$196.28	\$13.86	26%	100%

* The higher average price for bulk buyers is due to the larger bag size, not a preference for premium coffee.

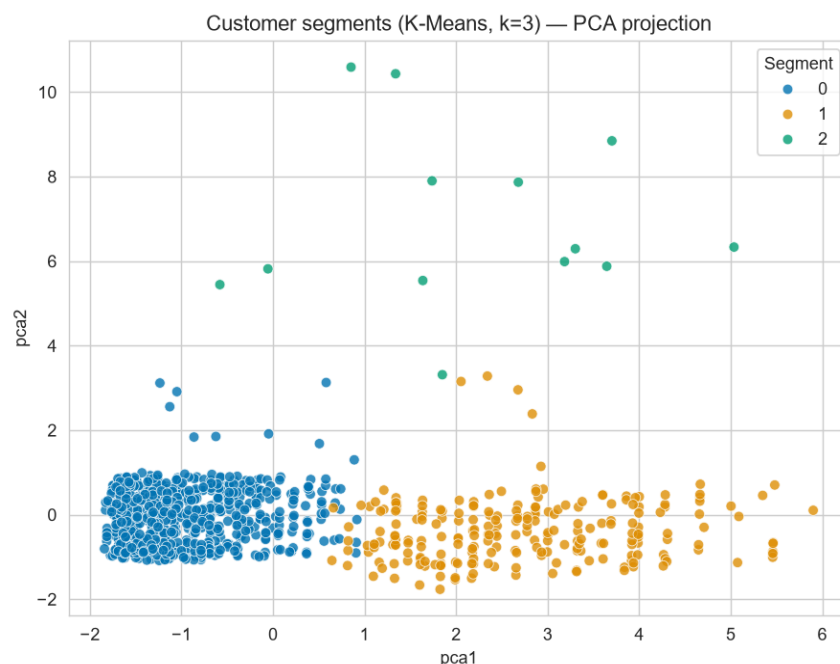


Figure 6. Customer segments (K-Means, k=3), PCA projection.

The high-value customer group included 13 of the 25 repeat customers. The remaining repeat customers were spread across the other two groups, showing that customer spending patterns alone cannot identify every repeat buyer. This suggests that additional information could improve customer segmentation.

6.2 Repeat-Purchase Classification

Leakage check: the model only used information available at a customer's first purchase. Features included coffee type, roast, size, unit price, quantity, order value, country, and loyalty card status. Information from future purchases, such as total spending or total number of orders, was excluded because it would unfairly influence the predictions.

The dataset was highly imbalanced, with only 2.7% of customers becoming repeat buyers. As a result, a model that predicts every customer will not return already achieves about 97% accuracy. This means accuracy alone is not a reliable measure of model performance, and other evaluation metrics are needed to assess how well the model identifies repeat customers.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Baseline (always “no repeat”)	97.4%	—	0%	—	0.50
Logistic Regression (class-weighted)	57.2%	2.1%	33.3%	3.9%	0.51
Random Forest (class-weighted)	93.0%	0%	0%	0%	0.52

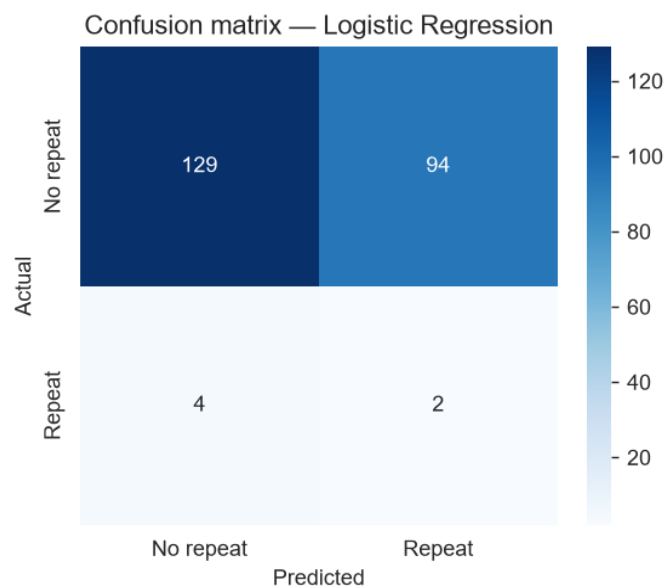


Figure 7. Confusion matrix, Logistic Regression (best F1 of the two models tested).

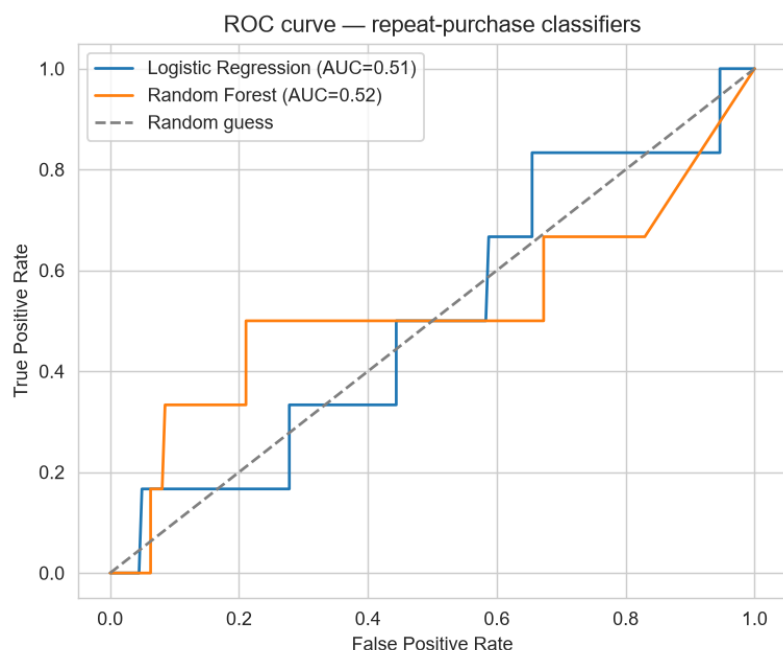


Figure 8. ROC curves — both models sit only marginally above the random-guess diagonal.

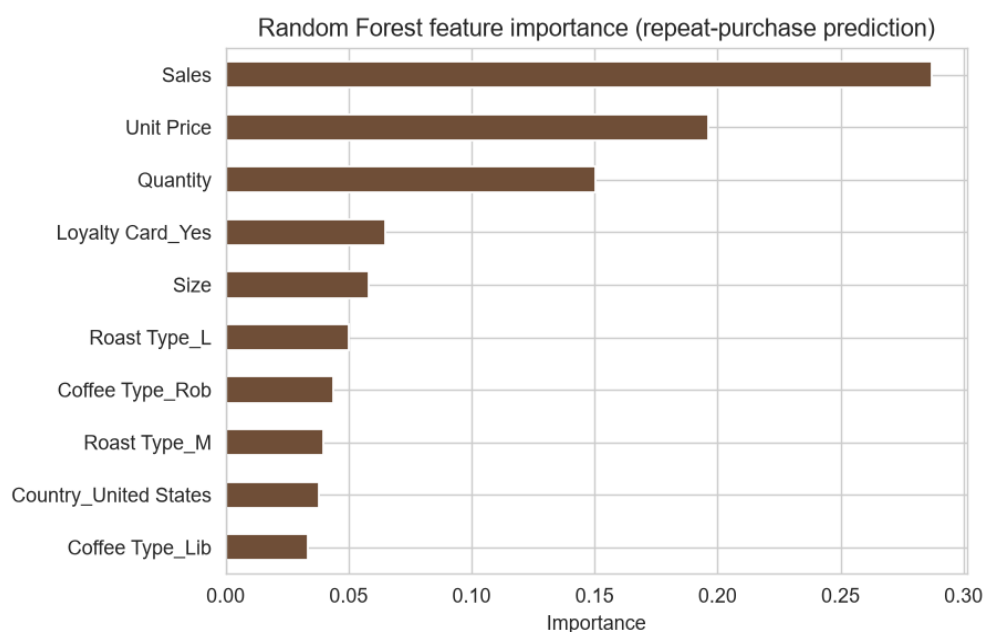


Figure 9. Random Forest feature importance for repeat-purchase prediction.

Both models performed only slightly better than random guessing (ROC-AUC of 0.51 and 0.52). This is largely because very few customers became repeat buyers, making it difficult for the models to learn meaningful patterns. The results suggest that the current dataset does not contain enough information to reliably predict repeat purchases. For this reason, customer segmentation produced more useful insights than classification, highlighting the need for additional customer behavior data in future work.

7. Interactive Dashboard

An interactive dashboard was developed using Plotly and exported as a standalone HTML file (dashboard/coffee_sales_dashboard.html). The dashboard allows users to explore sales trends, customer behavior, product performance, and key business metrics through interactive charts that can be opened directly in any web browser.

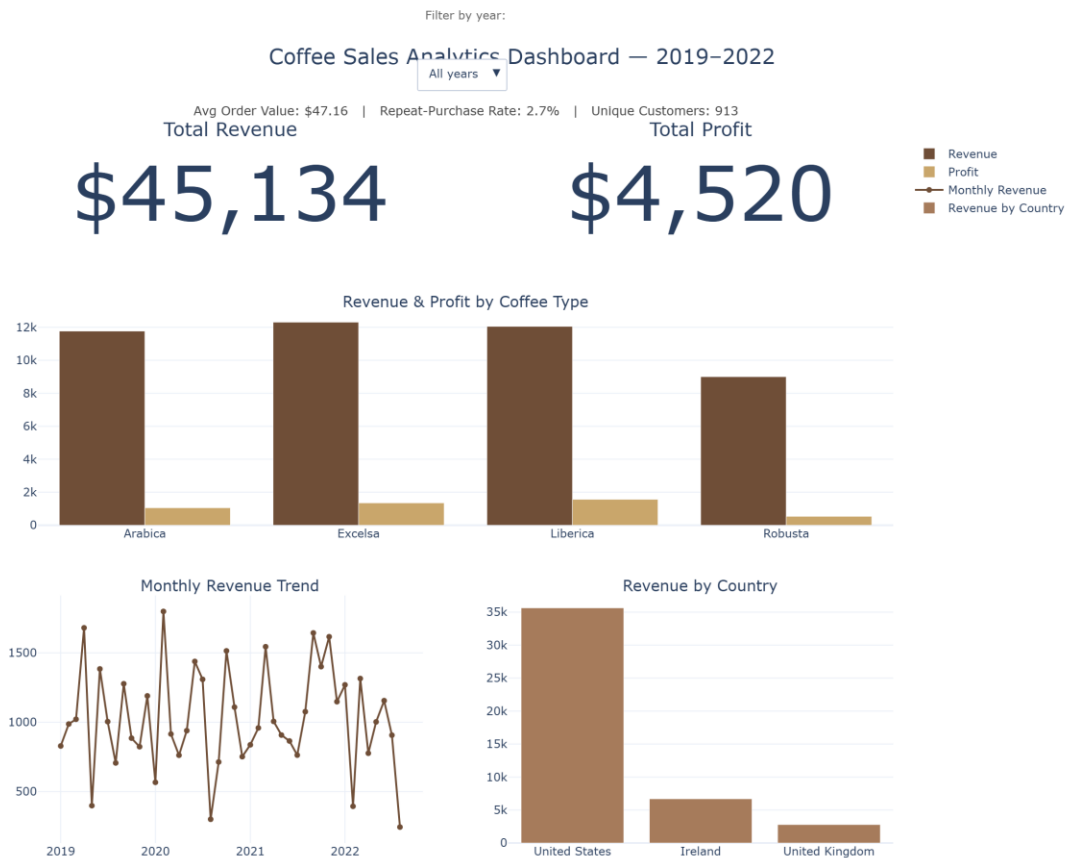


Figure 10. Interactive dashboard: total revenue, total profit, average order value, repeat-purchase rate, unique customers, revenue and profit by coffee type, monthly revenue trend, and revenue by country. The year dropdown filters every chart and KPI live.

KPIs shown: The dashboard presents key business metrics, including total revenue, total profit, average order value, repeat purchase rate, and total customers. It also includes interactive visualizations of revenue and profit by coffee type, monthly sales trends, and revenue by country, with a year filter for easy exploration.

8. Results Summary

- **Excelsa** generated the highest revenue (\$12,306), while **Liberica** had the highest profit margin (13%).
- The **2.5 kg** coffee size accounted for **52.7%** of total revenue, making it the top-selling product size.

- Only **25 out of 913 customers (2.7%)** made a repeat purchase, indicating a low customer retention rate.
- Customers enrolled in the loyalty program did not spend more than non-members.
- Sales peaked in **June** and were lowest in **August**. The 2022 sales trend also suggests slower growth compared to 2021.
- The **United Kingdom** recorded the lowest revenue and spending per customer, suggesting an opportunity to improve customer engagement.
- Customer segmentation identified **three distinct groups**: occasional buyers, bulk buyers, and high-value repeat customers.
- The repeat purchase prediction model performed only slightly better than random guessing, showing that the current dataset does not contain enough information to reliably predict repeat purchases.

9. Limitations

- The dataset is relatively small (1,000 order lines, 913 customers), limiting the scope of the analysis.
- Only 25 customers made repeat purchases, making it difficult to build a reliable prediction model.
- Customer demographic information was not available.
- Marketing data, such as promotions and email engagement, was not included.
- External factors such as weather, holidays, and economic conditions were not available for analysis.
- The findings are based on data from a single coffee retailer and may not generalize to other businesses.

10. Future Work

- Collect additional customer behavioral data such as marketing engagement to improve repeat purchase prediction.
- Connect the dashboard to a live database for real-time reporting.
- Develop a 12-month sales forecasting model to support inventory and business planning.
- Include customer acquisition cost and customer lifetime value to evaluate marketing effectiveness.
- Evaluate and improve the loyalty program to encourage repeat purchases.
- Retrain the prediction model using a larger dataset with more repeat customers.

11. Responsible AI

11.1 Data Privacy

The raw workbook includes customer name, email, phone number, and postal address. None of these are needed for sales, retention, or segmentation analysis, so they were dropped before any cleaned file

was written to this public repository (notebooks/01_data_cleaning.ipynb) — only the anonymous Customer ID is retained to join across tables. This is to protect customer privacy while preserving the data needed for analysis.

11.2 Bias and Fairness

The dataset itself is geographically imbalanced (79% of revenue from the U.S.); therefore, the findings and machine learning models are likely to represent U.S. customer behavior better than UK/Ireland customers. This limitation should be disclosed to any stakeholder using these models.

11.3 Transparency

The machine learning models were reported exactly as they performed. Since neither model reliably predicted repeat purchases, the results are presented as a limitation of the available data rather than overstating the model's performance.

11.4 Responsible Deployment

Were this deployed in production, the customer segmentation output should inform marketing and customer engagement strategies. However, the repeat purchase prediction model should not be used for business decisions until it is trained on a larger, more balanced dataset.

12. Conclusion

This project transformed raw coffee sales data into a complete data analytics workflow using Excel, SQL, Python, and an interactive Plotly dashboard. The analysis identified key business insights, including the highest-performing products, customer spending patterns, sales trends, and opportunities to improve customer retention.

Customer segmentation provided meaningful insights into different purchasing behaviors, while the repeat purchase prediction model highlighted the limitations of the available data. The project demonstrates the importance of understanding both the strengths and limitations of data when making business decisions.